FACULTY OF INFORMATION TECHNOLOGY
UNIVERSITY OF MORATUWA

25th ANNIVERSARY

SME-GPT

BILINGUAL FINANCIAL DOCUMENT INTELLIGENCE

FIT EXPO 2026

01. INDUSTRY PROBLEM & PROPOSED SOLUTION

We help Sri Lankan SME owners understand their money by turning paper documents into answers, in Sinhala or English.

TODAY
- Paper files, lost bills
- Totals added by hand
- Most tools English-first

→

WITH SME-GPT
- Upload a photo or PDF
- Review what it extracted
- Ask in your own language

02. INNOVATION & UNIQUE FEATURES

Bilingual by design
Reads and answers in Sinhala and English

AI plans, then computer calculates
Every total is worked out exactly, never guessed

Answers show their work
Every figure traces to its source

Catches overcharges
Flags bills priced above the agreed order

03. DEVELOPMENT PROCESS

Requirements → Design → Build → Test → Evaluate

- **Apr 2026** — Project start
- **May 2026** — 40-question test
- **Jul 2026** — Order tracking
- **Aug 2026** — Live on VPS

Pivot: local Llama 3 was too slow, so we moved to a cloud LLM

04. TECHNOLOGY CHOICES

Next.js + React
Responsive bilingual app

FastAPI (Python)
Close to the AI libraries

Surya OCR
Strong on mixed-language

DeepSeek
Affordable AI language model

Postgres + pgvector
Records plus search

multilingual-e5
Sinhala-capable, runs on CPU

05. SYSTEM ARCHITECTURE & TECHNICAL HIGHLIGHT

Next.js web app ⇄ **FastAPI backend** ⇄ **OCR + cloud LLM**

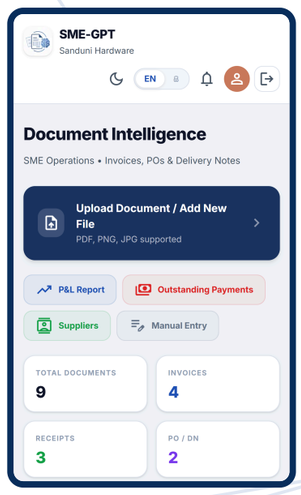
Upload: Preprocess → OCR → Correct → Extract → Review

Ask: Plan → **Validate** → **Compute** → Answer

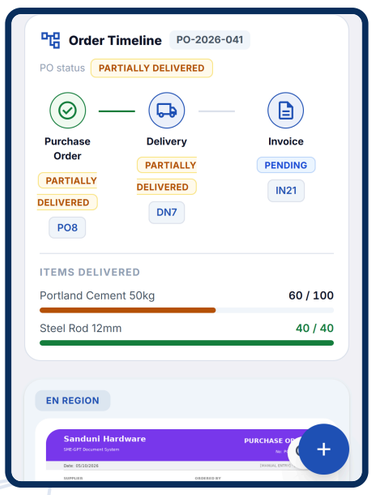
Secure database: every shop's records are kept private (pgvector + Postgres)

Challenge: AI can make maths mistakes, so it only plans the steps. A checker approves the plan and the computer does the maths.

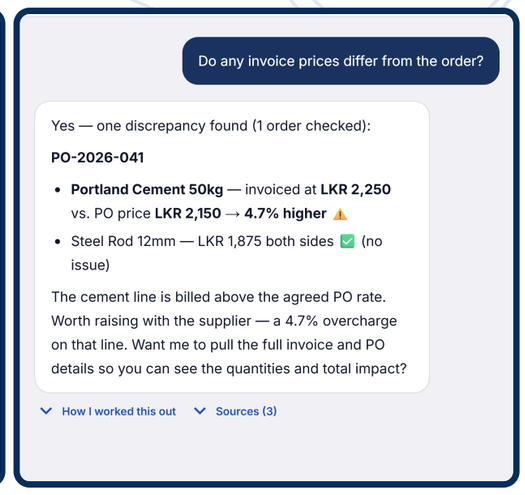
06. USER EXPERIENCE (UI/UX)



Dashboard



60 of 100 bags delivered



Overcharge caught

07. INDUSTRY VALIDATION & USER FEEDBACK

Correct answers in our 40-question test



Mr. Charith Bimsara
Reviewed our system design and recommended a cloud AI model.

Shop owners
A small group tried it.
Liked: full interaction in Sinhala
Disliked: less accuracy with low-quality inputs

08. PRACTICAL IMPACT, RESULTS & NEXT STEPS

IMPACT & RESULTS

540+	90%	~90–94%
automatic tests	answers correct	English bill details

Works now: live website in Sinhala and English, with order tracking

Benefit: faster answers, fewer billing mistakes

Limits: low-quality input (blurry photos), low-resource language (less Sinhala data)

09. WORKFLOW

1. Upload a document
2. AI reads it
3. Owner reviews, saves
4. Ask a question
5. See answer + source

NEXT STEPS

Future enhancements
Backup OCR engine and higher Sinhala accuracy

Development opportunities
Audited accuracy test and a larger SME trial

Industry collaborations
Partner with SME associations and accounting firms

Scan **Live app**

Scan **Source code**

SME-AI

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MEET THE TEAM

Scan to see our LinkedIn profiles and emails.